

The Informational Actualization Model (IAM):

Master Consolidated Summary

Current Status, Established Results, Validated Tests, and Complete Document Map

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March 3, 2026

Abstract

This document is the single consolidated master summary of the Informational Actualization Model (IAM) — a dual-sector cosmological framework derived from horizon thermodynamics. It records *(i)* what IAM claims and does not claim, *(ii)* what has been validated and by which pipelines, *(iii)* what each companion document contributes, and *(iv)* what remains for final submission polish.

The IAM cosmology core is a late-time framework in which photon-sector linear propagation remains standard ($\Sigma(a) = 1$), while matter-sector perturbations experience a late-time suppression ($\mu(a) < 1$). This suppression is activated by a fixed function $E(a) = \exp(1 - 1/a)$ and a baseline coupling $\beta_m \equiv \Omega_m/2$, derived from virial-partition arguments. The mechanism does *not* alter the background Friedmann expansion history — IAM is a perturbation-level modification only.

Validation has been organized into three complementary layers, executed across a total of **17 converged MCMC chains**: **(Level 1)** a μ - Σ implementation via MGCAMB/Cobaya using four Planck-era likelihood combinations plus a 7/7 Boltzmann-solver diagnostic suite; **(Level 2a)** direct dual-sector perturbation implementation in a CAMB-class Boltzmann solver via Fortran modifications (perturbations modified; background kept Λ CDM); and **(Level 2b)** exploratory background-modification diagnostics that converge but yield an unphysical low H_0 , confirming the viable IAM realization resides in the perturbation sector.

Two deterministic reproducibility scripts complement the MCMC campaigns: `iam_validation.py` produces a 10-point χ^2 breakdown and nine publication-quality figures; `iam_derivation_tests.py` runs a 15/15 derivation verification suite from Jacobson/Cai–Kim thermodynamic gravity through the IAM coupling predictions, including a zero-parameter test yielding $\Delta\chi^2 = 31.2$.

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1 What IAM Is — and Is Not

1.1 The Core Cosmology Claim

IAM is an effective, testable late-time modification to the gravitational sector at the level of linear perturbations. It adopts a horizon-thermodynamic/encoding viewpoint in the tradition of Jacobson [1995] and Cai & Kim [2005] and introduces one additional identification: irreversible classical information produced by gravitational structure formation contributes an informational term encoded on cosmic horizons. In the validated IAM realization, this term enters through the matter-sector perturbation equations as a sector-dependent effective expansion/friction, while null (photon) degrees of freedom remain completely unmodified at linear order.

The resulting observable signature is a late-time, scale-independent suppression of matter growth — parameterized as $\mu(a) < 1$ with $\Sigma(a) = 1$ — confrontable with CMB, RSD, lensing, and cluster observations using standard survey infrastructure.

1.2 What IAM Is Not Claiming

To prevent category errors common in referee review, the following non-claims are stated explicitly:

- IAM is not a replacement for the Standard Model of particle physics.
- IAM is not a complete UV theory of quantum gravity.
- IAM does not alter pre-recombination physics or the sound horizon in its viable realization.
- IAM does not require modifying the background Friedmann expansion equation.
- Particle-scale extensions (Koide ratio, electron rest mass) are *conditional derivations* with explicitly stated boundary assumptions, tracked separately from the empirically validated cosmology core.
- Black hole, quantum measurement, and decoherence papers are *extension tracks* and should not be conflated with the validated cosmology results unless independently tested.

Structural Note. The exploratory Level 2b runs explicitly tested the alternative hypothesis “modify the background expansion.” Those chains converge but yield an unphysical inferred $H_0 \approx 61.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$, confirming by elimination that the viable IAM realization is a **perturbation-sector modification**, not a global background change. This structural test was run intentionally as part of the validation program and rules out a common misreading of the model.

2 Minimal IAM Claim-Set (Cosmology Core)

2.1 The Three Core Inputs

The referee-relevant IAM cosmology core rests on three inputs, summarized in Table 1.

Table 1: The three referee-relevant inputs of the IAM cosmology core.		
#	Element	Statement
1	Sector separation	$\Sigma(a) = 1$ for photons; $\mu(a) < 1$ for matter (late-time activation).
2	Late-time activation	$E(a) = \exp\left(1 - \frac{1}{a}\right)$ — negligible at early times, $E(1) = 1$ today.
3	Baseline coupling	$\beta_m \equiv \frac{\Omega_m}{2}$ — motivated by the virial-partition argument; treated as a <i>derived input</i> , not a free fit parameter.

2.2 The μ - Σ Mapping

These three inputs map onto the standard (μ, Σ) modified-gravity parameterization used across the IAM cosmology papers:

$$\mu(a) = \frac{H_{\Lambda}^2(a)}{H_{\Lambda}^2(a) + \beta_m E(a)}, \quad (1)$$

$$\Sigma(a) = 1. \quad (2)$$

At $z = 0$, equation (1) yields $\mu(z=0) \approx 0.864$, i.e. $\mu_0 - 1 \approx -0.136$, a $\sim 14\%$ suppression of matter growth today. At high z , $E(a) \rightarrow 0$ and $\mu \rightarrow 1$, recovering Λ CDM at early times.

3 Validation Snapshot — Completed Work

IAM has been validated across three complementary levels using **17 total converged MCMC chains**. Each level addresses a distinct aspect of the model’s viability.

3.1 Level 1: MGCAMB/Cobaya μ - Σ Implementation

Level 1 validates the IAM fixed prediction $\mu_0 \simeq -0.135$ against Planck 2018 likelihoods and combinations with RSD, BAO, and Pantheon+ supernovae [Planck Collaboration, 2020]. Twelve chains were run across four dataset combinations. Across all combinations, $\Delta\chi^2$ remains below the 95% CL single-parameter threshold of 3.84, while σ_8 shifts downward relative to Λ CDM in the direction required by weak-lensing tensions. Table 2 summarizes the fixed- μ_0 comparison.

Table 2: Level 1 summary: fixed $\mu_0 = -0.135$ relative to Λ CDM. $\Delta\sigma_8 \equiv \sigma_8(\text{IAM}) - \sigma_8(\Lambda\text{CDM})$.

Dataset	$\Delta\chi^2$	$\sigma_8(\Lambda\text{CDM})$	$\sigma_8(\text{IAM})$	$\Delta\sigma_8$
Planck only	+1.43	0.8139	0.8014	-0.0125
Planck + RSD	+1.34	0.8131	0.8001	-0.0130
Planck + BAO	+2.32	0.8115	0.7981	-0.0134
Planck + Pantheon+	+1.58	0.8129	0.8000	-0.0129

3.1.1 Level 1 Boltzmann Solver Diagnostics: 7/7 Tests Passed

In addition to the MCMC runs, seven explicit Boltzmann-solver diagnostic tests verified the physical consistency of the Level 1 implementation; all seven pass (Table 3).

3.2 Level 2a: CAMB-Class Dual-Sector Perturbation Implementation

Level 2a implements the dual-sector mechanism directly in the perturbation equations of a CAMB-class Boltzmann code via Fortran modifications [Mahaffey, 2026c]. CDM and baryon perturbations evolve with an enhanced effective expansion rate; photon equations remain completely standard. The background Friedmann history is held fixed at Λ CDM throughout.

Table 3: MGCAMB Boltzmann validation (Level 1): 7/7 tests passed.

Test	Criterion	Result	Status
σ_8 range	In [0.79, 0.82]	0.7954	PASS
CMB TT ($\ell > 30$)	<1% residual	0.17%	PASS
CMB TT ISW ($\ell < 30$)	< cosmic variance	3.6% (CV $\sim 63\%$)	PASS
CMB lensing	<5% change	+0.30%	PASS
$\Sigma = 1$	Exact	$\Sigma_0 = 0$	PASS
$P(k)$	std(ratio) <1%	0.53%	PASS
scale-independence			
$f\sigma_8$ fit quality	$\chi^2 \leq \chi^2_{\Lambda\text{CDM}} + 4$	4.42 vs 4.85	PASS

Three chains were run: Run C (pure Λ CDM reference), Run A (IAM perturbation-only), and Run D (IAM + RSD data added). Under the full Planck CamSpec TTTEEE likelihood, the best-fit penalty is $\Delta\chi^2 = +0.54$ relative to Λ CDM — well within the noise floor. All standard parameters shift by $<0.1\sigma$. Results are collected in Table 4.

 Table 4: Level 2a side-by-side results (posterior mean $\pm 1\sigma$).

Parameter	Run C (Λ CDM)	Run A (IAM)	Run D (IAM+RSD)
H_0 [km/s/Mpc]	67.188 ± 0.465	67.161 ± 0.467	67.189 ± 0.460
σ_8	0.8087 ± 0.0059	0.7998 ± 0.0058	0.7995 ± 0.0058
Ω_m	0.3162 ± 0.0065	0.3166 ± 0.0065	0.3162 ± 0.0064
S_8	0.4547 ± 0.0062	0.4500 ± 0.0061	0.4496 ± 0.0060
Best χ^2	10,972.07	10,972.61	10,979.43
MCMC samples	115,584	81,088	53,312
$\Delta\chi^2$ (best-fit, Planck): IAM – Λ CDM = +0.54			

3.2.1 Dual-Sector H_0 Split (Posterior-Derived)

From the Level 2a Run A posterior, the matter-sector perturbation equations imply an effective matter-sector expansion rate that differs from the photon-sector rate, yielding:

$$H_0^{\text{matter}} = H_0 \sqrt{1 + \beta_m E(a=1)} = 67.161 \times \sqrt{1 + 0.15765} \approx 72.26 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad (3)$$

$$H_0^{\text{photon}} = 67.16 \text{ km s}^{-1} \text{ Mpc}^{-1}. \quad (4)$$

This split naturally spans the full Hubble tension range with *zero additional free parameters*. The photon-sector value is the standard Planck posterior; the matter-sector value is derived from the same posterior and the fixed coupling $\beta_m = \Omega_m/2$.

3.3 Level 2b: Exploratory Background-Modification Diagnostics

To eliminate the alternative hypothesis that the IAM mechanism should be implemented as a background expansion modification, two additional chains were run with modified Friedmann equations.

These chains converge excellently (acceptance rates $\sim 46\%$, $R - 1 < 0.01$), but yield an unphysical low inferred $H_0 \approx 61.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (Table 5). This structurally confirms that the perturbation-only realization is the viable IAM implementation.

Table 5: Level 2b background-modification diagnostics (exploratory). Chains converge but yield unphysical H_0 , ruling out background-level implementation.

Run	Accepted samples	Acceptance	$R - 1$
Run A (background modified)	125,440	46.8%	0.0096
Run D (background modified)	120,832	46.3%	0.0068

4 Deterministic Reproducibility Scripts

Two deterministic Python scripts provide fast, reproducible cross-checks of IAM’s quantitative predictions, independent of the full MCMC pipeline.

4.1 `iam_validation.py` — 10-Point Demonstration Dataset

This script operates on a compact demonstration dataset of 10 points (3 H_0 measurements + 7 $f\sigma_8$ growth measurements) and produces a full χ^2 breakdown plus nine publication-quality figures. It does not replace the Planck-era likelihood pipelines in Levels 1 and 2; rather, it provides a transparent, fast verification of IAM’s key quantitative claims. Results are given in Table 6.

Table 6: Deterministic validation summary from `iam_validation.py` (10 data points: 3 H_0 + 7 growth).

Test	Quantity	Result
1	Λ CDM baseline χ^2	$\chi_{H_0}^2 = 31.91$, $\chi_{\text{growth}}^2 = 6.36$, $\chi_{\text{total}}^2 = 38.28$
2	IAM dual-sector χ^2	$\chi_{H_0}^2 = 1.52$, $\chi_{\text{growth}}^2 = 6.75$, $\chi_{\text{total}}^2 = 8.27$
2	Improvement	$\Delta\chi_{\text{total}}^2 = 30.01$ (5.5σ); $\Delta\chi_{H_0}^2 = 30.40$
3	Profile likelihood	$\beta_m = 0.157 \pm 0.029$ (68%); $\beta_m = 0.157 \pm 0.058$ (95%)
4	Photon-sector (profile)	$\beta_\gamma < 0.004$ (95% CL)
8	Photon-sector (MCMC)	$\beta_\gamma < 1.40 \times 10^{-6}$ (95% CL); $\beta_\gamma/\beta_m < 8.50 \times 10^{-6}$
7	Model selection	$\Delta\text{AIC} = 26.01$; $\Delta\text{BIC} = 25.40$; $\mathcal{L}(\Lambda\text{CDM})/\mathcal{L}(\text{IAM}) = 2.25 \times 10^{-6}$

Script-generated figures (exact filenames as printed by the script):

- `figure1_h0_comparison.pdf`
- `figure2_growth_suppression.pdf`
- `figure3_growth_rate.pdf`

- `figure4_beta_gamma_constraint.pdf`
- `figure5_beta_m_profile.pdf`
- `figure6_h0_ladder_complete.pdf`
- `figure7_chi2_breakdown.pdf`
- `figure8_summary_panel.pdf`
- `figure9_mcmc_corner.pdf`

4.2 `iam_derivation_tests.py` — 15/15 Derivation Verification Suite

This script runs 15 programmatic verification tests tracing the full IAM derivation chain — from Jacobson/Cai–Kim thermodynamic gravity foundations through to the IAM activation function, coupling predictions, μ – Σ mapping, and robustness checks. All 15 tests pass (Table 7).

Table 7: Derivation verification suite: 15/15 PASS. All results are exact script output.

#	Topic	Key Printed Diagnostic
1	Jacobson (1995)	$E^2(a=1) = 1.0000000000$; Raychaudhuri match True
2	Cai–Kim (2005)	Flatness 1.0000000000 ; $T_H/H = 0.159155$
3	Modified entropy	$E_{\text{IAM}}^2(1) = 1 + \beta = 1.157500$; $E(0.01) \rightarrow 0$
4	Integral fit	Best $n = 3.5$: $\alpha = 2.283$, $\beta = 1.009$, $r = 0.643533$
5	Sheth–Tormen	Best $\sigma^* = 1.5$: $\beta = 1.058$
6	Virial theorem	$\beta_m = \Omega_m/2 = 0.1575$ matches MCMC to 0.3%
7	Collapse fraction	$f_{\text{coll}} = 0.5927$ supports virial vs. naive
8	μ , Σ	$\mu(0) = 0.8639$; $\Sigma = 1$
9	Zero-parameter test	$\Delta\chi^2 = 31.2$ (10-point demo set)
10	w_{info}	$w_{\text{eff}}(0) = -1.0623$; mapped w_0 , w_a
11	Continuity	Residual 7.60×10^{-16}
12	MGCAMB approximation	Max error 4.29 pp at $z = 0$
13	Exponent sensitivity	μ_0 spread 0; σ_8 spread 0.0031
14	Coupling sensitivity	Virial β near minimum-tension point
15	Not $w_0 w_a$ CDM	Distances degenerate; μ differs; growth breaks degeneracy

5 Established Results — Canonical Locked List

The following results are treated as established within the IAM cosmology core, having been defined, implemented, and validated by the completed Level 1 and Level 2 campaigns:

1. **Sector signature:** $\Sigma = 1$ for photons and $\mu(a) < 1$ for matter, with late-time activation. This is the fundamental, falsifiable IAM prediction.
2. **Level 1 viability:** fixed- μ_0 prediction yields $\Delta\chi^2 < 3.84$ across all four Planck-era dataset combinations (well below the 95% CL threshold).

3. **Level 1 Boltzmann validation:** 7/7 solver diagnostics pass, confirming physical consistency of the μ - Σ implementation.
4. **Level 2a Planck viability:** $\Delta\chi^2 = +0.54$ best-fit relative to Λ CDM with all standard parameters stable ($<0.1\sigma$) and σ_8 suppressed from 0.8087 to 0.7998.
5. **Perturbation-only structure confirmed:** background-modification diagnostics converge but yield unphysical $H_0 \approx 61.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$, supporting the perturbation-only realization by elimination.
6. **Dual-sector H_0 split:** $H_0^{\text{matter}} = 72.26 \text{ km s}^{-1} \text{ Mpc}^{-1}$ inferred alongside $H_0^{\text{photon}} = 67.16$, naturally spanning the Hubble tension with zero free parameters.
7. **Cluster discriminator:** $M_{\text{lens}}/M_{\text{dyn}} = 1/\mu(z)$ with a distinctive redshift trend toward unity at high z , qualitatively distinct from redshift-independent hydrostatic bias.
8. **Zero-parameter test:** $\Delta\chi^2 = 31.2$ on the 10-point demonstration dataset with zero free parameters (5.5σ improvement over Λ CDM on that dataset).

6 Complete IAM Document Map

The IAM project comprises a network of documents covering the validated cosmology core, testable predictions, extension tracks, and reproducibility infrastructure. Each entry below gives the filename followed by a brief description of its purpose and value. Status: **core** = validated cosmology; **ext** = conditional extension track; **opt** = optional appendix.

Core cosmology papers

- **IAM.Manuscript.pdf [core]**
Master narrative; the single canonical “what IAM is / is not.” Integrates all other components and serves as the primary referee entry point.
- **IAM.Theory.Paper.pdf [core]**
Core derivation spine: derives the $E(a)$ activation function, the μ - Σ sector-split, and the coupling $\beta_m = \Omega_m/2$. First-principles foundation for all cosmology tests.
- **Late.Time.Growth.Suppression.(mu-Sigma).pdf [core]**
Level 1 cosmology test: confronts the fixed prediction $\mu_0 = -0.135$ with Planck 2018 and LSS across four dataset combinations. Establishes viability ($\Delta\chi^2 < 3.84$ throughout) and the direction of the σ_8 shift.
- **Dual.Sector.Perturbation.Cosmology.CAMB.pdf [core]**
Level 2a: direct dual-sector perturbation implementation in CAMB-class solver via Fortran modification. Demonstrates perturbation-only Planck viability and the H_0 sector split with zero free parameters.
- **IAM.CAMB.Technical.Note.pdf [core]**
Implementation details, convergence diagnostics, and the Level 2b background-modification exploratory outcomes.
- **Dual.Sector.Validation.Paper.pdf [core]**
Validation framing and cross-check notes; records consistency logic and cross-configuration sanity checks.
- **IAM.Test.Validation.Compendium.pdf [core]**
Central test ledger with the full validation ladder and all headline scores; one-stop reference for all validation results.
- **Supplementary.Methods.Reproducibility.Guide.pdf [core]**

End-to-end reproduction steps, run-level inputs/outputs, and tolerances; ensures any collaborator can reproduce all results from scratch.

- [IAM_Technical_Clarifications_Guide.pdf](#) [core]
Reader-defense document: explicit definitions, scope boundaries, and non-claims to prevent misinterpretation of the validated core.

Observational predictions and falsification

- [IAM_Lensing_Dynamics_Paper.pdf](#) [core]
Derives the cluster discriminator $M_{\text{lens}}/M_{\text{dyn}} = 1/\mu(z)$ for the $\Sigma = 1$ case; provides the clean redshift-dependent falsification signature.
- [3Way_Mass_Discrepancy_in_Galaxy_Clusters.pdf](#) [core]
SZ/X-ray/lensing ordering tests and internal consistency relations; separates IAM from pure hydrostatic bias explanations.
- [IAM_Survey_Predictions_Paper.pdf](#) [core]
Forecast and falsification roadmap for Euclid-class datasets; pre-registers IAM’s near-term decision points.

Coupling motivation and nonlinear closure

- [Virial_Efficiency_and_Effective_Nonlinear_Exponent.pdf](#) [core]
N-body literature consistency closure; shows the required virial efficiency and effective nonlinear exponent are consistent with published simulation analyses, strengthening the “derived parameter” narrative for $\beta_m = \Omega_m/2$.
- [Virial_Partition_Across_Wide_Domains.pdf](#) [core]
Cross-domain partition argument motivating $\beta_m = \Omega_m/2$ as a mathematical property of $1/r$ potentials.
- [IAM_M_Sigma_Paper.pdf](#) [core]
Connects IAM logic to M – σ style scaling; bridge between cosmological and astrophysical structure scaling.

Extension tracks (conditional; separate from cosmology validation)

- [IAM_BH_Thermodynamics.pdf](#) [ext]
Black hole thermodynamic extension of the horizon-encoding logic; clarifies what “encoding” means at BH horizons.
- [IAM_Black_Hole_Information_Paradox.pdf](#) [ext]
Applies IAM information bookkeeping to the BH information paradox.
- [Gravitational_Decoherence_Quantum_Level.pdf](#) [ext]
Proposes lab-scale decoherence tests bridging the encoding logic to measurable rates; extension track with specific experimental proposals.
- [IAM_Measurement_Problem_Quantum.pdf](#) [ext]
Frames the quantum measurement problem in IAM language; explicitly separated from the cosmology validation.
- [Electron_Rest_Mass_from_IAM.pdf](#) [ext]
Derives an electron mass relation under stated boundary/encoding assumptions; kept segregated from cosmology validation.

- `Koide_ratio_from_holographic_boundary.pdf` [ext]
Conditional theorem: given a boundary-encoding ansatz plus virial amplitude constraint, the Koide ratio follows algebraically.

Reproducibility scripts and optional appendices

- `iam_validation.py` [core]
Deterministic demonstration script: 10-point χ^2 breakdown plus nine publication-quality figures; fast reproducible cross-check of key quantitative claims.
- `iam_derivation_tests.py` [core]
15/15 derivation verification suite: automated check of the full IAM derivation chain from Jacobson/Cai–Kim foundations through the coupling predictions.
- `wz_far_future.pdf` [opt]
Optional appendix: asymptotic behavior and far-future extrapolation under IAM activation logic; no Big Rip under the baseline model.

7 Falsification Program and Survey Forecasts

IAM is explicitly designed to be falsified by near-term survey data. The following tests are pre-registered as the model’s primary near-term decision points.

7.1 The Cluster Lensing-to-Dynamics Discriminator

Because $\Sigma = 1$ (photons unmodified) but $\mu(z) < 1$ (matter suppressed), IAM predicts

$$\frac{M_{\text{lens}}}{M_{\text{dyn}}} = \frac{1}{\mu(z)}. \quad (5)$$

The key discriminating feature is the *redshift dependence*: this ratio approaches unity at high z as $E(a) \rightarrow 0$, producing a trend qualitatively distinct from roughly constant hydrostatic bias explanations. The predicted three-way mass ordering is $M_{\text{lens}} > M_{\text{SZ}} \approx M_{\text{hydro}}$, since both SZ and hydrostatic estimators are matter-sector and are suppressed identically.

7.2 Growth Suppression in RSD and $f\sigma_8$

IAM predicts suppressed $f\sigma_8(z)$ relative to Λ CDM, concentrated at $z \lesssim 2$. The prediction is a smooth, monotonic suppression of growth directly comparable against any $f\sigma_8$ measurement at known redshift, with no additional free parameters beyond the derived coupling β_m .

7.3 Euclid-Class Detectability and Falsification Criteria

- **Conservative falsification criterion:** if μ_0 is constrained to > -0.05 at 95% CL by Euclid-class data [Frusciante et al., 2025], the IAM baseline is excluded.
- **ISW sign:** IAM predicts a modified ISW amplitude distinguishable from Λ CDM with next-generation CMB-LSS cross-correlation data.
- **CMB lensing:** the $\Sigma = 1$ prediction implies unchanged lensing power; any deviation at the IAM-predicted σ_8 level is a strong tension with the model.

8 Remaining Work — Completion Checklist

The following items remain; the list intentionally excludes work already completed in the MCMC chains, preprints, and scripts. The remaining work is primarily **consolidation and presentation**, not re-running completed tests.

8.1 For This Master Summary Paper

- Freeze a single IAM Baseline Parameter Set (Planck values, β_m computation, μ_0 , σ_8 targets) and reference it consistently across all documents.
- Finalize and standardize the flagship figure suite (6–10 figures) and insert final PDFs.
- Add a one-page “Probe Sector Assignment” box listing which observables are primarily photon/geometry vs. growth/potential, to prevent misinterpretation.
- Final notation polish: unify Level 1 μ - Σ presentation with Level 2 perturbation-only solver language (particularly μ_0 sign conventions).
- Add a single “Reproducibility Pointer” with one canonical entry: methods guide + compendium.

8.2 For Cosmology Validation (Next Concrete Deliverables)

- Produce one unified summary table: Planck-only, Planck+RSD, Planck+BAO, Planck+SNe with consistent notation throughout.
- Generate the primary cluster redshift test plot: $M_{\text{lens}}/M_{\text{dyn}}$ vs. z with IAM prediction overlay and at least one compiled dataset comparison.
- Pre-register a quantitative falsification threshold for Euclid-class constraints on μ_0 .
- Integrate one “predicted band vs. literature points” plot for the virial-efficiency consistency closure paper.

8.3 For Extended Physics Papers (Optional / Separate Track)

- **Koide:** add a robustness appendix showing sensitivity to small perturbations of the boundary-encoding ansatz; keep explicitly conditional.
- **Electron mass:** add assumption sensitivity analysis and numerical stability notes.
- **Quantum decoherence:** connect to named experimental regimes with a benchmark table (mass, temperature, predicted τ ranges).

9 New Session Bootstrap Paragraph

The following paragraph is the canonical seed statement for beginning any new discussion of the IAM project.

IAM Quick Reference. IAM (Informational Actualization Model) is a dual-sector cosmology derived from horizon thermodynamics. Its core, validated claim-set is: matter perturbations experience a late-time suppression $\mu(a) < 1$ while photons obey $\Sigma(a) = 1$ exactly. The predicted form is $\mu(a) = H_\Lambda^2(a)/[H_\Lambda^2(a) + \beta_m E(a)]$, with $\beta_m = \Omega_m/2$ and $E(a) = \exp(1 - 1/a)$. This has been validated for Planck 2018 consistency via both μ - Σ (Level 1, 12 chains) and explicit dual-sector CAMB implementations (Level 2a, 3 chains), producing $\sigma_8 \approx 0.800$ and a matter/photon H_0 split

(72.26 vs 67.16 $\text{km s}^{-1} \text{Mpc}^{-1}$) without modifying the ΛCDM background. Key falsifiable predictions: $M_{\text{lens}}/M_{\text{dyn}} = 1/\mu(z)$ with a distinctive redshift trend, and near-term survey detectability of $\mu_0 \approx -0.136$ by Euclid-class data. Particle-scale and quantum extensions (Koide, electron mass, decoherence) are tracked as an independent conditional track and should not be conflated with the validated cosmology core.

Acknowledgments

The author thanks the developers of MGCAMB, Cobaya, and CAMB for public code access.

Data Availability

MCMC chains and analysis scripts are available at:
<https://github.com/hmahaffeyges/IAM-Validation>

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